

Full Stack Java Developer Roadmap (With Spring Boot)

■ Phase 1: Core Java Foundation (Weeks 1–4)

■ Goal: Build strong Java fundamentals and object-oriented programming skills.

- 1 Java Basics: JDK, JRE, JVM, Data Types, Operators, Control Flow
- 2 OOP Concepts: Classes, Objects, Inheritance, Polymorphism, Encapsulation, Abstraction
- 3 Exception Handling: try-catch-finally, throw/throws, custom exceptions
- 4 Collections Framework: List, Set, Map, Queue
- 5 Multithreading: Thread class, Runnable interface, Synchronization
- 6 Java 8+ Features: Lambda Expressions, Streams, Functional Interfaces
- 7 File I/O & Serialization

■ Project: Console-based Employee Management System using Java Collections.

■ Phase 2: Front-End Basics (Weeks 5–8)

■ Goal: Build a strong understanding of front-end web fundamentals.

- 1 HTML5: Structure, forms, tables, multimedia, semantic tags
- 2 CSS3: Styling, Flexbox, Grid, Responsive design, Animations
- 3 JavaScript (ES6+): Variables, Functions, Loops, Objects, Arrays, DOM Manipulation, Fetch API

■ Project: Personal portfolio or landing page using HTML, CSS, and JS.

■ Phase 3: Advanced Front-End (Weeks 9–12)

■ Goal: Learn modern front-end frameworks and connect with backend APIs.

- 1 React.js: Components, Hooks, Router, State Management (Context API / Redux), API Integration
- 2 Angular (Alternative): Components, Directives, Modules, Services, Routing, HTTP Client

■ Project: To-Do App or Weather App integrated with a mock API.

■ Phase 4: Back-End Development (Weeks 13–16)

■ Goal: Master Java backend development with Spring Boot.

- 1 Spring Core: Dependency Injection, Beans, Autowiring, Annotations
- 2 Spring Boot: Auto Configuration, Application Properties, Starter Dependencies

- 3 RESTful APIs: @RestController, @GetMapping, @PostMapping
- 4 Spring Data JPA & Hibernate: Entities, Repositories, Relationships, CRUD operations
- 5 Spring Security & JWT Authentication: Authentication, Authorization, Role-based access
- 6 Validation & Exception Handling
- 7 Spring Boot Testing with JUnit & Mockito

■ Project: User Management REST API using Spring Boot + MySQL + JWT.

■ Phase 5: Database Skills (Weeks 17–18)

■ Goal: Learn SQL and NoSQL databases and integrate with Spring Boot.

- 1 SQL (MySQL/PostgreSQL): CRUD, Joins, Transactions, Views, Indexes
- 2 NoSQL (MongoDB): Documents, CRUD, Aggregation, Spring Data MongoDB

■ Project: Integrate Spring Boot REST API with MySQL and MongoDB.

■ Phase 6: DevOps & Tools (Weeks 19–20)

■ Goal: Learn essential tools for building, testing, and deploying applications.

- 1 Git & GitHub: Version control, branching, pull requests
- 2 Maven / Gradle: Build automation and dependency management
- 3 Jenkins: CI/CD basics
- 4 Docker: Containerize Spring Boot + Frontend apps
- 5 Postman: Test and debug APIs
- 6 Deployment: Host Spring Boot API on AWS EC2 / Render, Frontend on Vercel or Netlify

■ Project: CI/CD pipeline with GitHub → Jenkins → Docker → AWS.

■ Phase 7: Final Full Stack Project (Weeks 21–22)

■ Goal: Combine everything into a real-world full stack app.

- 1 E-Commerce Website (Products, Cart, Orders, JWT Auth)
- 2 Job Portal App (Users, Recruiters, Applications)
- 3 Task Management System (Tasks, Teams, Comments)
- 4 Blogging Platform (CRUD posts, JWT Auth, MongoDB backend)

■ Tech Stack: React.js/Angular + Spring Boot + MySQL/MongoDB + JWT + Docker + AWS